

Muhammmad Athar Abbas
SP25-BSE-082
Object Oriented Programming
Theory Assignment 4

Main Class:

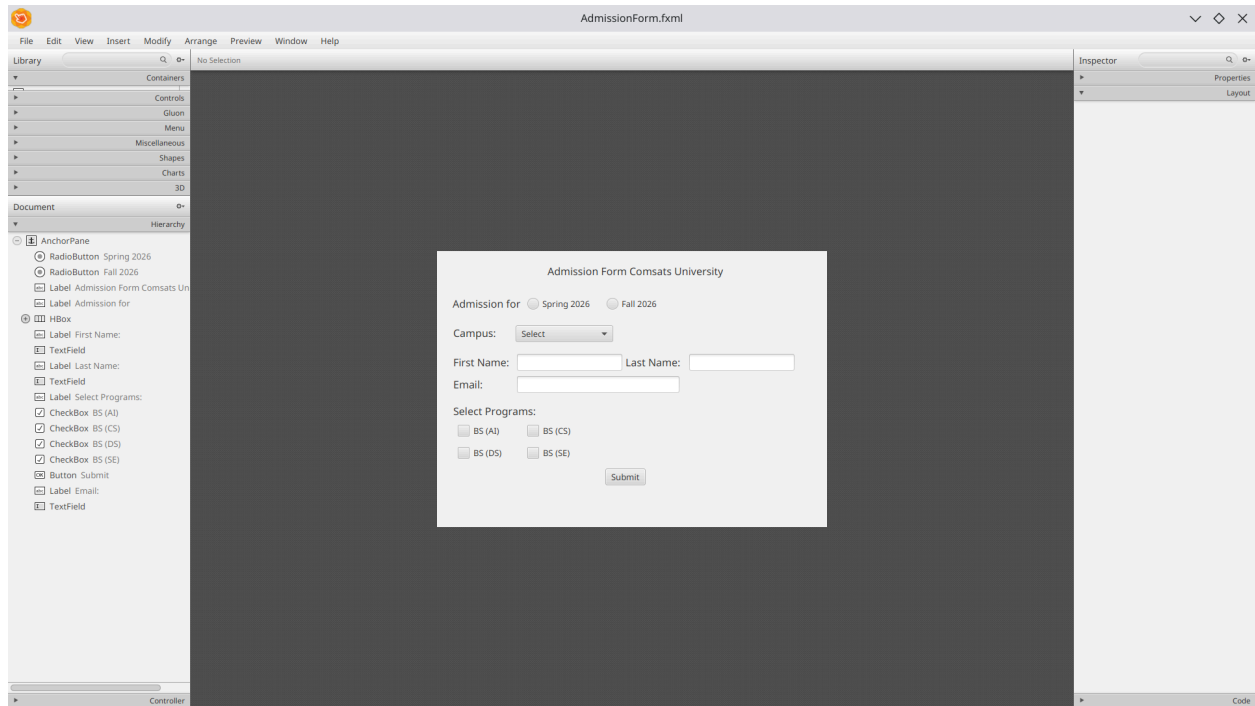
```
package application.admissionform;

import javafx.application.Application;
import javafx.fxml.FXMLLoader;
import javafx.scene.Parent;
import javafx.scene.Scene;
import javafx.stage.Stage;

public class Main extends Application {

    @Override
    public void start(Stage stage) throws Exception {
        FXMLLoader loader =new
FXMLLoader(getClass().getResource("AdmissionForm.fxml"));
        Parent root = loader.load();
        FormController controller = loader.getController();
        controller.initliaze();
        Scene scene = new Scene(root);
        stage.setScene(scene);
        stage.show();
    }
    public static void main(String[] args){
        launch(args);
    }
}
```

Gluon SceneBuilder:



FXML File:

```
<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.CheckBox?>
<?import javafx.scene.control.ComboBox?>
<?import javafx.scene.control.Label?>
<?import javafx.scene.control.RadioButton?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.control.ToggleGroup?>
<?import javafx.scene.layout.AnchorPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.text.Font?>

<AnchorPane prefHeight="424.0" prefWidth="600.0"
xmlns="http://javafx.com/javafx/25"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="application.admissionform.FormController">
```

```

    <children>
      <RadioButton layoutX="139.0" layoutY="72.0"
mnemonicParsing="false" text="Spring 2026">
        <toggleGroup>
          <ToggleGroup fx:id="adminrad" />
        </toggleGroup>
      </RadioButton>
      <RadioButton layoutX="261.0" layoutY="72.0"
mnemonicParsing="false" text="Fall 2026" toggleGroup="$adminrad" />
      <Label layoutX="170.0" layoutY="8.0" prefHeight="46.0"
prefWidth="281.0" text="Admission Form Comsats University">
        <font>
          <Font size="16.0" />
        </font>
      </Label>
      <Label layoutX="24.0" layoutY="58.0" prefHeight="46.0"
prefWidth="112.0" text="Admission for">
        <font>
          <Font size="16.0" />
        </font>
      </Label>
      <HBox layoutX="25.0" layoutY="104.0" prefHeight="46.0"
prefWidth="326.0">
        <children>
          <Label prefHeight="43.0" prefWidth="84.0" text="Campus:">
            <font>
              <Font size="16.0" />
            </font>
          </Label>
          <ComboBox fx:id="campusesCombo" prefWidth="150.0"
promptText="Select">
            <HBox.margin>
              <Insets left="12.0" top="10.0" />
            </HBox.margin>
          </ComboBox>
        </children>
      </HBox>

```

```

    <Label layoutX="25.0" layoutY="158.0" prefHeight="26.0"
prefWidth="98.0" text="First Name:">
        <font>
            <Font size="16.0" />
        </font>
    </Label>
    <TextField layoutX="123.0" layoutY="158.0" />
    <Label layoutX="290.0" layoutY="158.0" prefHeight="26.0"
prefWidth="98.0" text="Last Name:">
        <font>
            <Font size="16.0" />
        </font>
    </Label>
    <TextField layoutX="388.0" layoutY="158.0" />
    <Label layoutX="25.0" layoutY="229.0" prefHeight="34.0"
prefWidth="144.0" text="Select Programs:">
        <font>
            <Font size="16.0" />
        </font>
    </Label>
    <CheckBox layoutX="32.0" layoutY="263.0" mnemonicParsing="false"
prefHeight="26.0" prefWidth="84.0" text="BS (AI)" />
    <CheckBox layoutX="139.0" layoutY="263.0" mnemonicParsing="false"
prefHeight="26.0" prefWidth="84.0" text="BS (CS)" />
    <CheckBox layoutX="32.0" layoutY="297.0" mnemonicParsing="false"
prefHeight="26.0" prefWidth="84.0" text="BS (DS)" />
    <CheckBox layoutX="139.0" layoutY="297.0" mnemonicParsing="false"
prefHeight="26.0" prefWidth="84.0" text="BS (SE)" />
    <Button layoutX="259.0" layoutY="334.0" mnemonicParsing="false"
text="Submit" />
    <Label layoutX="25.0" layoutY="194.0" text="Email:">
        <font>
            <Font size="16.0" />
        </font>
    </Label>
    <TextField layoutX="123.0" layoutY="192.0" prefHeight="22.0"
prefWidth="250.0" />
</children> </AnchorPane>

```

FXML Controller:

```
package application.admissionform;

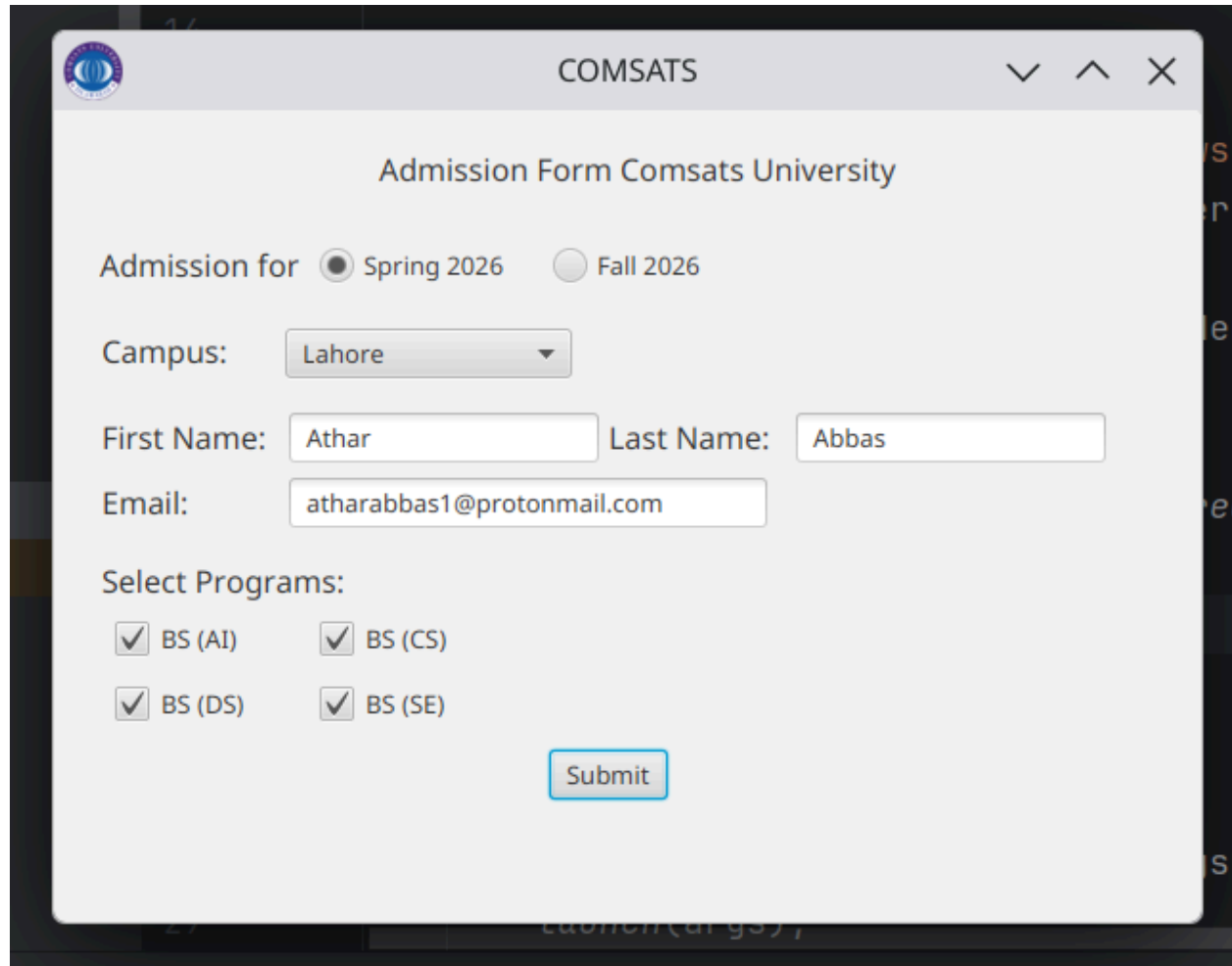
import javafx.fxml.FXML;
import javafx.scene.control.ComboBox;

import static javafx.collections.FXCollections.observableArrayList;

public class FormController {
    @FXML
    private ComboBox<String> campusesCombo;
    public void initliaze(){
        campusesCombo.setItems(observableArrayList("Lahore",
        "Islamabad", "Attock", "Abottabad","Sahiwal", "Vehari"));
    }
}
```

(output on next page)

Output:



The image shows a web browser window titled "COMSATS" displaying the "Admission Form Comsats University". The form is for the "Spring 2026" session. The user has selected "Lahore" as the campus, entered "Athar" as the first name, "Abbas" as the last name, and "atharabbas1@protonmail.com" as the email address. Under "Select Programs:", all four options are checked: BS (AI), BS (CS), BS (DS), and BS (SE). A "Submit" button is located at the bottom right of the form.

COMSATS

Admission Form Comsats University

Admission for ☒ Spring 2026 ☐ Fall 2026

Campus:

First Name: Last Name:

Email:

Select Programs:

☒ BS (AI) ☒ BS (CS)

☒ BS (DS) ☒ BS (SE)